

Photoelectric effect supports the particulate nature of EM waves.

37. Answer: A

$$\Delta E = hf$$

$$= \frac{hc}{\lambda}$$

$$= \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{633 \times 10^{-9}}$$

$$= 3.142 \times 10^{-19} \text{ J}$$

$$= 1.96 \text{ eV}$$

Only $W \rightarrow X$ has an energy difference of 1.96 eV.